INTRODUCTION

3.1 OVERVIEW OF METABOLISM

We have already learnt in XI standard about the structures and functions of various biomolecules such as proteins, amino acids, carbohydrates etc... We know that biomolecules are important for the normal functioning of any living system. These biomolecules can be obtained either directly from the diet or synthesised in cells. In the next few chapters, we will study the synthesis of various biomolecules and their degradation. These biochemical studies are termed as metabolism. Metabolism consists of series of connected enzymatic reactions that produce specific products. The different species involved in these reactions are referred to as metabolites. Photosynthesis is an example of carbohydrate metabolism in which a series of chemical reactions convert carbon dioxide and water to glucose. In the next few chapters, we will study about the metabolism of various biomolecules such as proteins, lipds etc. Figure 3.1 gives the overview of the various metabolic processes.

We learnt in the previous unit, that digestion of carbohydrates produce three major products namely glucose, galactose and fructose. These monosaccharides are absorbed into the blood stream. After their absorption, monosaccharides are transported to the liver, where fructose and galactose are rapidly converted into glucose.

In this unit we will study the metabolism of carbohydrates. Carbohydrates are simpler when compared to other biomolecules such as nucleic acids, amino acids etc... and their metabolism supplies more than half of the energy requirements of the body. In fact, our brain mainly depends upon carbohydrate metabolism as a source of energy.

3.1.1 Catabolism and anabolism

Every cell contains thousands of reactions comprising many metabolic pathways. There are two types of metabolic reactions namely anabolism and catabolism. The process of synthesising the required larger biomolecules from the small molecules is called anabolism. These reactions use up energy for example, in photosynthesis, small molecules CO_2 and water are converted into a six carbon carbohydrate. In this reaction sun light provides the energy.

$$6CO_2 + 6H_2O + 2800kJ \xrightarrow[Chlorophyll\&Enzymes]{Sunlight} C_6H_{12}O_6 + 6O_2$$

Catabolism is a reverse process of anabolism, in which bigger biomolecules are broken into simple molecules. Let us consider the oxidation of glucose,

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 2800kJ$$

The above reaction is the reversal of photosynthesis reaction in which large (6C glucose) molecules are broken into small $CO_2$and H_2O molecules. If the glucose is burnt in the lab we get CO_2 & water and large amount of energy is released. Whereas, in living cell, the oxidation of glucose proceeds through a series of enzyme-catalysed intermediate reactions and the part of the energy produced in these reactions is stored in the chemical bonds of ATP (Adenosine triphosphate). When energy is needed, for example during the muscle contraction, ATP is utilised.

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